The Journal of Open Source Software

MetaNLP: Natural Language Processing for Meta Analysis

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 27 August 2025
Published: unpublished

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Summary

The time-consuming task of title-abstract screening in the preparation of a systematic review can be partially automated through the use of machine learning models. These models are often trained on the document-term matrix derived from the collection of titles and abstracts. In this paper, we present the R package **MetaNLP**, which allows the transfer of a CSV file containing titles and abstracts into a document-term matrix with just a few lines of code. Furthermore, it provides a variety of functions to modify this matrix according to user preferences and includes helpful summary and visualization features.

Statement of need

In the process of conducting a systematic review, title and abstract (TIAB) screening is especially time-consuming, as it is standard practice for at least two human reviewers to independently assess all identified citations and determine whether each should be included or excluded from the subsequent full-text screening. However, this task involves plain text reading and could potentially be performed using natural language processing (NLP) and machine learning (ML) methods. A certain proportion of the citations to be screened can be classified by humans and used as training data to train a ML model. This ML model can then classify the remaining citations, either acting alone or alongside a second human reviewer.

Besides pre-implemented, but non-flexible ML tools like Rayyan (Ouzzani et al., 2016) or fine-tuned large language models (Dennstädt et al., 2024), there are approaches that take the titles and abstracts of the publications, compute the document-term matrix and train ML models based on this matrix (Kebede et al., 2023; Lange et al., 2021; Pilz et al., 2024). This approach provides the user with the flexibility to exclude negligible words and to train different ML algorithms, making it easier to apply the best algorithm for the problem at hand.

The R-package **MetaNLP** supports programmers that aim to employ this approach. It provides all necessary functionality to create and flexibly modify the document-term matrix from a CSV file that contains titles and abstracts. Unlike most general-purpose NLP packages, **MetaNLP** is designed to be part of a pipeline to train ML models, and allows users to have text preprocessing, customizable feature construction and data visualization methods unified in a single package.

Methodology

The titles and abstracts must be organized in a CSV file, with each row representing one citation. The CSV must have a column `ID` to clearly identify each citation, a column `title`

containing the titles, and another column `abstract`. A decision column is only required when the CSV is used to train supervised ML algorithms.

MetaNLP then processes the CSV in several distinct steps: At first, the CSV file containing the titles and abstracts is read and scanned for missing values or empty strings. Words containing special characters or numbers are removed, upper case letters are set to lower case, and all unnecessary empty spaces are stripped.

In the second step, each word is lemmatized. Here, we make use of the package **textstem** (Rinker, 2018).

Following lemmatization, the words are subjected to stemming. The algorithm conducting the stemming is called Porter's stemming algorithm (Porter, 1980), implemented via the R-package **SnowballC** (Bouchet-Valat, 2023).

In the final step, the document-term matrix is created. For each lemmatized and stemmed word, the number of appearances in the title and abstract is determined, potentially weighted, and written into the matrix.

The whole processing pipeline described above makes use of the R-package **tm** (Feinerer & Hornik, 2024).

Additionally, **MetaNLP** offers the function `select_features` to apply elastic net regularization within the data processing workflow. Here, the R-package **glmnet** (Friedman et al., 2010; Tay et al., 2023) is applied.

Usage

MetaNLP creates the document-term matrix from a CSV file containing all the abstracts and the "include/exclude" decision. It should have the following form:

ID	decision	title	abstract
1	exclude	Postoperative	A degree of feed intolerance after neona
2	exclude	Selective decc	Gastrointestinal surgery is associated wi
3	include	Perioperative	Infectious complications and especially a
4	include	Randomized c	Infectious complications and anastomoti

The creation of the document-term matrix is simple now.

```
library(MetaNLP)
dtm <- MetaNLP("data/example.csv",
               bounds = c(2, Inf),
               word_length = c(3, Inf),
               weighting = "frequency",
               language = "english")
```

The function `MetaNLP()` creates an object of type `MetaNLP` with a slot `data_frame`, which is the document-term matrix, and this object is then passed on to the other functions.

Visualize `MetaNLP` objects

The function `summary` provides a comprehensive overview of the data, showing the most common words in total and stratified by "include" and "exclude".

```
summary(dtm, n = 5, stop_words = TRUE)
```

One way to visualize a `MetaNLP` object is the `plot` function, which creates a bar chart of the n most common words.

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```
plot(dtm, n = 10, decision = "total", stop_words = FALSE)
```

Delete words

A typical step to reduce the number of columns is the deletion of stop words.

```
dtm_delete <- delete_stop_words(dtm)
```

In general, an arbitrary list of words can be easily removed using the method `delete_words`.

```
dtm_delete <- delete_words(dtm_delete, c("aim", "administer"))
```

Dimension reduction

In order to further reduce the dimensionality of the feature space, **MetaNLP** offers the method `select_features`.

```
dtm_1se <- select_features(dtm, alpha = 0.2, lambda = "1se", seed = 42)
```

The argument `alpha` denotes the elastic net mixing parameter and `lambda` is the parameter of the penalty.

Test data

Assume a ML model has been trained on the document-term matrix and will now be used to classify additional citations. For this, the document-term matrix of the new (test) data must have exactly the same features as the matrix of the training data. The function `read_test_data` takes a `MetaNLP` object and the path to the test data. It then assimilates its columns by removing columns that only appear in the test data and by adding columns as zero-columns that only exist in the document-term matrix used for training.

```
test_dtm <- read_test_data(dtm, "data/example_test.csv")
```

Acknowledgements

The authors have not received any funding for this work.

References

Bouchet-Valat, M. (2023). *SnowballC: Snowball stemmers based on the c 'libstemmer' UTF-8 library*. https://CRAN.R-project.org/package=SnowballC

Dennstädt, F., Zink, J., Putora, P. M., Hastings, J., & Cihoric, N. (2024). Title and abstract screening for literature reviews using large language models: An exploratory study in the biomedical domain. *Systematic Reviews*, *13*(1), 158. https://doi.org/10.1186/s13643-024-02575-4

Feinerer, I., & Hornik, K. (2024). *Tm: Text mining package*. https://CRAN.R-project.org/package=tm

Friedman, J., Tibshirani, R., & Hastie, T. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, *33*(1), 1–22. https://doi.org/10.18637/jss.v033.i01

Kebede, M., Le Cornet, C., & Fortner, R. (2023). In-depth evaluation of machine learning methods for semi-automating article screening in a systematic review of mechanistic literature. *Research Synthesis Methods*, *14*(2), 156–172. https://doi.org/10.1002/jrsm.1589

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Lange, T., Schwarzer, G., Datzmann, T., & Binder, H. (2021). Machine learning for identifying relevant publications in updates of systematic reviews of diagnostic test studies. *Research Synthesis Methods*, *12*(4), 506–515. https://doi.org/10.1002/JRSM.1486

Ouzzani, M., Hammady, H., Fedorowicz, Z., & Elmagarmid, A. (2016). Rayyan — a web and mobile app for systematic reviews. *Systematic Reviews*, *5*(210). https://doi.org/10.1186/s13643-016-0384-4

Pilz, M., Zimmermann, S., Friedrichs, J., Wördehoff, E., Ronellenfitsch, U., Kieser, M., & Vey, J. A. (2024). Semi-automated title-abstract screening using natural language processing and machine learning. *Systematic Reviews*, *13*(1), 274. https://doi.org/10.1186/s13643-024-02688-w

Porter, M. (1980). An algorithm for suffix stripping. *Program*, *14*(3), 130–137. https://doi.org/10.1108/eb046814

Rinker, T. W. (2018). *textstem: Tools for stemming and lemmatizing text*. http://github.com/trinker/textstem

Tay, J. K., Narasimhan, B., & Hastie, T. (2023). Elastic net regularization paths for all generalized linear models. *Journal of Statistical Software*, *106*(1), 1–31. https://doi.org/10.18637/jss.v106.i01

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